

EECT CE 6325 VLSI Design

Fall 2026

PROJECT #1: Verilog/VHDL

Due: **Wed September 9**

Project Introduction

You are to create an arbitrary digital design of your choice using Verilog or VHDL. Behavioral Verilog is the highest level of abstraction without details for hardware implementation. You will not have to define gates in your code; we will have a program later in Project 2 to translate your behavioral Verilog code into structural Verilog code. Designing at this level is similar to C programming.

The only restriction is that the design must use flip-flops. Examples include FIR and IIR filters, finite state machines, etc. You can pick a design of your choice as long the above restrictions are met. After project 2 your design needs to use 3000 cells or more. However, at this stage you will not know the cell count. So it may be wise to choose a scalable design, one that is easy to extend, in case it is too small after the synthesis project.

This design does not have to be used for your final project, but with a little planning now, you will not have to create a new design at the end of the semester. If you make your design scalable, you will not have to redo your design if it synthesizes to a small design. Reviewing what Verilog code is synthesizable will make your life much easier when you get to Project 2.

No credit will be given for any design copied from the internet or from other sources.

Project Description/Requirements

- 1) We want you to code your design in Behavioral Verilog/VHDL.
- 2) One test bench and at least one module for your FSM.
- 3) Clear comments to your code including comments in your test bench that make it clear how you are testing your module(s).

Report Layout

1. A cover page containing student names (max of 3), NetID and project title.
2. A paragraph on the general description of your digital design.
3. A block diagram of the module(s). Please label all inputs and outputs.
4. A block diagram of how the module(s) and test bench are connected.
5. A state diagram of your design with data flow.
6. A copy of your simulation waveform results with enough comments to make it clear that your design functions correctly.
7. A soft copy of your code.
8. A soft copy of project report is required, upload the document on Canvas (**Projects -> Project 1 -> Project 1 Report**).

Project Flow and Tutorial Links (cited Tutorials are in the tutorial folder on Canvas)

- 1) You will setup NX NoMachine client and be familiar with the environment. (1NomachineTutorial)
- 2) You will simulate your Verilog code. (HDL-Verilog - VLSI Tutorial)